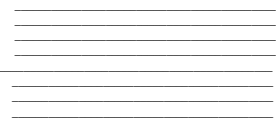


Developing Intelligent Alarms to Predict Critical Deterioration for
Hypotensive Events using Multi-Source Physiological Data

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Abstract

Alarm fatigue in intensive care units (ICUs) is a critical patient safety issue caused by excessive, non-actionable alerts from static threshold-based monitoring systems that fail to reflect individual physiological variability. While existing research has largely focused on false alarm suppression or threshold customization, few have explored predictive alarm systems capable of anticipating clinical deterioration. This thesis presents a machine learning-based approach to forecast hypotensive events within a five-minute horizon using high-resolution, multi-source physiological waveform data. Leveraging over 19,500 hours of data from The Hospital for Sick Children (SickKids), models were trained on arterial blood pressure, pulse oximeter, and ECG signals. Two-layer gated recurrent unit (GRU) networks outperformed convolutional neural networks (CNNs), achieving test accuracy of ~92%, recall over 97%, and precision around 87%—a significant improvement over traditional systems with false alarm rates as high as 99%. These results indicate the model’s strong capacity to detect clinically meaningful events while minimizing missed detections and reducing false positives. The findings demonstrate the feasibility and value of shifting from reactive to proactive alarm systems in critical care, offering earlier, more actionable warnings, improving clinical response times, and reducing caregiver burden. This research provides a validated framework for intelligent, predictive monitoring systems and represents a meaningful step toward precision-based healthcare in high-acuity environments.

Keywords: Alarm Fatigue, Hypotension, Machine Learning, Intelligent Alarm Systems, Multivariate Time Series, Physiological Waveform Analysis

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1 Introduction

In intensive critical care, patients are continuously monitored through various devices that track vital signs such as heart rate, blood pressure, electrocardiograms (ECGs), respiratory function, and more. These monitoring systems are designed to detect abnormalities by utilizing basic algorithms that trigger alarms when physiological parameters deviate from predefined "normal" ranges. These thresholds are generally determined by broad population categories, such as age and weight. However, these static and generalized parameters often fail to capture the unique and dynamic physiological conditions of individual patients. As a result, critical care units are inundated with frequent non-actionable alarms, with caregivers experiencing between 150 and 400 alarms per patient per day. Alarming, studies have shown that 85–99% of these alarms are either false positives or clinically insignificant [1].

The overabundance of alarms has led to the widespread phenomenon of "alarm fatigue," a condition in which caregivers become desensitized to alarms due to their excessive frequency and lack of actionable relevance [2]. This desensitization undermines trust in alarm systems and interferes with the timely and effective care of patients. As one study notes, "ICU nurses think that alarms are burdensome and too frequent, interfering with caring for patients and causing reduced trust in alarm systems" [3]. This underscores a critical flaw in current alarm systems: their inability to adapt to the specific and evolving physiological characteristics of individual patients. Consequently, the intended purpose of these systems—to provide timely and actionable alerts—is often compromised.

While prior studies have explored intelligent alarm systems through false alarm suppression or multivariate data integration, few have shifted the paradigm from reactive to predictive alarms. Most solutions focus on real-time alarm classification or simple threshold customization, without leveraging the predictive potential of machine learning over continuous physiological waveforms. This reveals a key research gap: the lack of alarm systems that can proactively predict critical deterioration, rather than respond to it after it has occurred.

This research addresses this pressing problem in critical care: the high frequency of non-actionable alarms that contribute to alarm fatigue and compromise patient safety. The central objective of this research is to develop an intelligent alarm system that can predict critical deterioration for hypotensive events by analyzing multi-source physiological data using machine learning techniques. We hypothesize that a predictive algorithm that adapts according to this information will significantly reduce non-actionable low blood pressure alarms while providing earlier, actionable warnings for clinical intervention, thereby curbing the widespread issue of alarm fatigue.

To achieve this, we train and evaluate deep learning models using retrospective waveform data from The Hospital for Sick Children (SickKids), focusing on a 5-minute prediction horizon. By integrating arterial blood pressure, pulse oximeter, and ECG waveforms, this system learns temporal dependencies and multivariate interactions in patient data. The project also compares recurrent and convolutional architectures to evaluate performance and generalizability, with an emphasis on real-time applicability in intensive care environments.

Beyond alleviating the burden on clinicians, the proposed system has the potential to significantly improve early intervention and clinical outcomes for patients. Furthermore, this work could pave the way for broader applications of intelligent monitoring systems in other areas of healthcare. By advancing precision medicine approaches in acute care settings, the study aspires to contribute to the ongoing transformation of healthcare into a more personalized, efficient, and effective domain.

2 Background

This chapter aims to provide the background for my thesis through a comprehensive literature review, covering the motivation and necessity for intelligent alarm systems, current approaches taken towards smart alarms using multivariate physiological data integration and false alarm classification with machine learning, and finally identifying and bridging the gaps in current research that my research aims to address.

2.1 Necessity for Intelligent Alarm Algorithms

The motivation and necessity for intelligent alarm systems in intensive critical care is supported by a growing body of research highlighting significant shortcomings and problems with current alarm systems. These systems, primarily designed around generic, population-based thresholds, empirically fail to adequately accommodate the unique physiological profiles of individual patients, leading to an excessive number of non-actionable or clinically insignificant alarms. Alarm fatigue, a condition in which caregivers become desensitized to alarms due to their excessive frequency and lack of actionable relevance, is a direct consequence of such frequent and often unnecessary alerts. Alarm fatigue has been widely documented as a critical safety concern in intensive care units (ICUs), undermining patient safety and clinician effectiveness.

Lewandowska et al. [3] systematically reviewed the impact of alarm fatigue in ICU settings, revealing that nurses perceive alarms as burdensome, overly frequent, and disruptive to patient care. The study highlighted alarm fatigue as a primary factor contributing to diminished trust in alarm systems, decreased patient care quality, and delayed responses to potentially critical situations, ultimately compromising patient safety. Nurses reported being overwhelmed by continuous false alarms, prompting coping mechanisms such as alarm silencing or ignoring, further amplifying safety risks.

Supporting these findings, Sowan et al. [4] reported that despite interventions such as adjusting default alarm settings and educational programs,

alarm fatigue persisted due to the inherently high rate of alarms generated by cardiac monitors. Their results indicated that simple adjustments to alarm thresholds were insufficient, as alarms continued to inundate clinical staff, highlighting the complexity of effectively managing alarm systems without more sophisticated solutions.

Similarly, Cho et al. [5] identified that nearly 80% of nurses viewed nuisance alarms as frequent interruptions negatively impacting their trust in alarm systems. The lack of differentiation between alarm priorities and unclear alarm sources exacerbated confusion, delaying critical interventions. Nurses emphasized the necessity of clearly distinguishing alarm priorities and sources to enhance clinical responsiveness and reliability.

The potential effectiveness of customized alarm thresholds to mitigate alarm fatigue is propagated by studies emphasizing patient-specific alarm settings. According to research on customizing alarm limits based on specific patient needs, a disproportionate number of alarms often originate from a small subset of patients, highlighting a significant shortcoming of current generic presets [6]. Customizing thresholds based on patient-specific baselines dramatically reduced unnecessary alarm triggers, such as adjusting low heart rate thresholds for patients with chronically low baselines due to conditions like atrial fibrillation.

Moreover, unit-specific variations in alarm frequencies and types further emphasize the inadequacy of manufacturer-provided default settings, necessitating presets that reflect unit-specific populations, such as ICUs, telemetry, or pediatric units. The most frequent non-actionable alarms, including threshold breaches in parameters such as heart rate (HR), blood pressure (BP), respiratory rate (RR), and peripheral capillary oxygen saturation (SpO_2) [3], further justify the requirement for patient-tailored configurations.

The recognition of these critical issues in literature motivates the development of intelligent alarm systems. Such systems would assess whether or not an alarm should be triggered for a specific patient based on real-time analyses of multi-source physiological data, patient history, and individual clinical

conditions, effectively addressing the identified gaps. By reducing false positives, intelligent alarms aim to significantly minimize alarm fatigue and enhance clinical trust and responsiveness, ultimately improving patient safety and outcomes. Thus, the transition from current static, reactive alarm mechanisms to proactive, adaptive predictive systems emerges as both a clinical necessity and an opportunity to revolutionize patient care in critical care settings.

2.2 Multivariate Physiological Data Integration

Integrating multiple physiological signals has emerged as a critical strategy for enhancing the accuracy and specificity of alarm systems in intensive care units (ICUs). The current literature strongly supports the benefit of multivariate data integration, demonstrating substantial improvements over traditional single-parameter threshold-based alarm systems.

The study by Stevens et al. ("Smart Alarms: Multivariate Medical Alarm Integration for Post-CABG Surgery Patients") [7] introduced the effectiveness of integrating multiple vital signs such as heart rate (HR), blood pressure (BP), respiratory rate (RR), and oxygen saturation (SpO₂). By employing a fuzzy expert system complemented by a Bayesian clinical decision support tool, their approach significantly reduced false alarms by an average of 59%, thereby enhancing clinical decision-making and intervention timing. This integration allows for identification of clinically significant patterns and interactions between different physiological parameters, overcoming limitations inherent to single-parameter monitoring systems.

Hever et al. [8] further validated the potential of multi-sensor integration through machine learning approaches. Their research demonstrated that applying random forest techniques to multi-sensor ICU data significantly reduced the false alarm rate even in cases where sensor data was incomplete or intermittently missing. The use of multi-sensor fusion enabled robust alarm accuracy by leveraging correlated physiological signals, ensuring that the reliability of critical warnings remains intact even in lossy, less-than-ideal monitoring conditions.

Kilinc and Ghattas [9] proposed an intelligent alarm system that employs fuzzy logic and multi-parameter validation algorithms. Their system notably included 23 algorithms designed to cross-check multiple parameters such as HR, BP, and SpO₂. The incorporation of delays and multiparameter validation effectively distinguished transient, clinically insignificant anomalies from sustained, critical conditions requiring intervention. By iteratively developing these algorithms through clinician focus groups, their system significantly reduced alarm fatigue, further illustrating the advantage of multi-parameter validation in capturing patient-specific physiological deviations.

Ordoñez et al. [10] introduced novel multivariate methods, such as Multivariate Bag-of-Patterns (BoPs) and Stacked BoPs, tailored for physiological time series data classification. These methods efficiently captured complex physiological interactions across multiple signals across a variety of patients, significantly improving the capability to classify clinically relevant physiological states such as hypotension or cardiac events. This research emphasizes the value of sophisticated data representation techniques to enhance clinical decision support and alarm accuracy through effective handling and analysis of multivariate physiological datasets.

Together, these studies collectively affirm that integrating multiple physiological signals enhances alarm specificity, reduces non-actionable alerts, and supports more accurate and precise clinical interventions. My research builds upon these findings as it utilizes multiple physiological signals in the form of time series data/waveforms with advanced machine learning techniques, leveraging this proven enhanced alarm accuracy and precision to reduce false negatives, thereby curbing the issue of alarm fatigue. By shifting the focus from reactive to predictive alerting mechanisms, this approach also aims to deliver earlier, more precise warnings, significantly improving clinical outcomes and patient safety in intensive care environments.

2.3 False Alarm Classification using Machine Learning

Machine learning (ML) has emerged as a powerful approach for improving alarm accuracy and precision in intensive care units (ICUs), specifically to addressing the overarching problem of alarm fatigue caused by frequent false alarms. Several studies demonstrate the efficacy of ML algorithms in distinguishing true alarms from non-actionable alerts, significantly reducing cognitive load and improving clinical monitoring quality.

Ansari et al.'s "Suppression of False Arrhythmia Alarms in the ICU: A Machine Learning Approach" [11] developed a multimodal detection algorithm integrating multiple waveforms such as ECG, arterial blood pressure (ABP), and photoplethysmogram (PPG). The study leveraged various beat detection algorithms whose outputs were combined using machine learning techniques. Specifically, separate classification models for ventricular tachycardia (VT) and ventricular fibrillation (VF) were trained to discriminate between normal and abnormal beats effectively, leading to substantial false alarm suppression without significant loss of sensitivity. This integration presents the potential of ML in enhancing alarm accuracy and specificity by incorporating detailed waveform analyses and processing of multi-source physiological signals.

Moreover, Wang et al. [12] developed a comprehensive ML-based approach to ICU false alarm detection, extracting 216 relevant features from both arterial blood pressure and electrocardiogram signals. Unlike previous methods requiring specialized algorithms for different alarm types, their generalized multivariate approach successfully distinguished true alarms from false ones, demonstrating robust performance across various arrhythmia conditions. This method achieved high classification accuracy, underscoring ML's capability to handle complex, multivariate data scenarios prevalent in critical care environments.

As previously mentioned, Hever et al. [8] emphasized the strength of ML models in handling missing sensor data, a frequent issue in ICU settings. By employing random forest (RF) classifiers trained on ICU multi-sensor data, the study demonstrated significant reductions in false alarms, even with deliberately

omitted parameters such as heart rate or blood pressure. Their results showed that RF models maintained high sensitivity and positive predictive values despite missing data, effectively compensating through the integration of available sensor information. This robustness to incomplete data represents a significant advantage of ML over traditional, parameter-specific algorithms.

Au-Yeung et al. [13] introduced an optimized ML approach combining signal processing, feature extraction, and advanced random forest algorithms to reduce false arrhythmic alarms. The method involved sophisticated extraction of both physiological and signal quality features, which significantly improved discrimination between true and false alarms. Their system achieved the highest score reported in literature for real-time alarm classification, clearly demonstrating ML's ability to enhance alarm precision and reduce false positives effectively.

Similarly, Eerikäinen et al. [14] developed an ML algorithm specifically for classifying true and false alarms of five life-threatening arrhythmias using random forests. This approach integrated physiological and signal quality features derived from ECG, PPG, and ABP signals, achieving high classification sensitivity and specificity. This robust performance emphasizes the potential for ML algorithms to dynamically adapt and accurately classify complex physiological data in real-time scenarios.

Lastly, the work by Bizzego et al. [15] even further demonstrated the efficacy of deep learning and transfer learning in processing multivariate physiological signals. Their approach leveraged deep neural networks (DNNs) to classify different types of physiological signals, significantly improving pattern recognition and diagnostic capabilities. This research highlights how advanced machine learning techniques, particularly deep learning, can extract valuable insights from complex multivariate datasets even without feature extraction or engineering, significantly surpassing traditional, univariate analytical methods.

These studies collectively demonstrate the strength of ML techniques in significantly improving alarm precision and accuracy in critical care, attributed to the ability to integrate multiple physiological signals to capture complex patterns in

various patient archetypes. Most of these existing approaches focus on real-time false alarm classification and rely on feature extraction for immediate discrimination of true from false alarms. My research aims to extend this capability further by adding predictive capabilities, while employing the many evident strengths of ML techniques for improved alarm accuracy and precision. Specifically, my approach employs machine learning techniques to proactively predict critical hypotensive deteriorations precisely and accurately using continuous, multi-source physiological data. By shifting from a purely reactive alarm classification model to a proactive predictive model, this research seeks to further reduce alarm fatigue, enhance clinical intervention timing, and improve patient safety outcomes in ICUs.

2.4 Forecasting Patient Conditions within Prediction Horizons

Machine learning (ML) offers a significant advantage over traditional threshold-based alarm systems by leveraging historical data to predict future patient conditions as opposed to single-parameter real-time alarms, providing earlier but still actionable alerts for critical deteriorations. Several studies have demonstrated the feasibility and accuracy of predictive ML models across various medical contexts, affirming their potential in enhancing patient monitoring and clinical outcomes.

Simon et al. [16] explored the predictive accuracy of machine learning algorithms over different time horizons for incident myocardial infarction (MI) using longitudinal data. The study demonstrated that ML models, particularly L1 regularized (Lasso) regression and random forests, achieved high predictive accuracy at moderate follow-up periods (approximately 1500 to 2500 days). Compared to traditional methods like the Framingham Heart Study model, ML algorithms provided superior performance in predicting MI events, showcasing their ability to effectively handle multivariate data and non-linear relationships between patient features and clinical outcomes. These findings reflect the potential of ML models in delivering accurate classifications within time horizons, thus enhancing preventive clinical interventions.

Loh et al. [17] also demonstrated the value of prediction horizons in clinical ML applications by comparing binary classification models with time-to-event (TTE) methods for diagnosing conditions such as autism ADHD. The study highlighted significant limitations of traditional binary classification approaches in scenarios characterized by inconsistent patient follow-up, which often result in biased outcomes due to censoring effects. Conversely, discrete-time neural networks (DTNN), a TTE approach, demonstrated superior ability to handle right-censoring by accurately predicting event probabilities over longer horizons. This improved predictive performance illustrates how TTE models effectively capture the temporal nature of clinical data, providing more precise forecasts of patient deterioration events compared to traditional classification methods.

Predictive machine learning techniques have been successfully employed in intensive care unit (ICU) settings to anticipate critical events on a shorter timeframe. In studies such as one by Hyland et al. [18], ML algorithms utilizing longitudinal physiological data, including vital signs, laboratory values, and clinical notes, effectively predicted deterioration episodes in critically ill patients. These models consistently outperformed traditional monitoring systems by offering earlier warnings with higher specificity and sensitivity, significantly improving clinical decision-making capabilities and potentially reducing morbidity and mortality rates.

Lastly, Ordoñez et al. [10] applied ML-based forecasting methods using multivariate time series analyses of physiological data. They introduced sophisticated data representations (multivariate bag-of-patterns), which captured complex interdependencies amongst physiological signals. These methods demonstrated superior classification accuracy in predicting acute clinical deteriorations, indicating that predictive models have the potential to significantly outperform conventional threshold-based systems by providing timely alerts for clinical intervention.

Collectively, these studies demonstrate not only the feasibility of predictive ML approaches but also their robustness and clinical utility in various healthcare

settings. By effectively using historical physiological data, predictive ML models surpass traditional reactive alarm systems by anticipating critical events, thereby enabling earlier and more targeted clinical interventions. My research leverages this predictive potential, specifically targeting the forecast of hypotensive events using multi-source physiological data. By shifting alarm systems from reactive to proactive while improving alarm accuracy, my approach aims to enhance clinical responsiveness, significantly reduce alarm fatigue, and improve patient outcomes in critical care environments.

2.5 Identifying and Bridging Gaps in Current Research

The preceding review of existing literature has highlighted several critical gaps in the current design and operation of clinical alarm systems, which are the pillars justifying my research approach.

Firstly, existing alarm systems primarily rely on static, generalized thresholds that fail to account for individual patient variability and dynamic physiological conditions, resulting in high rates of non-actionable alarms. As section 2.1 demonstrated, the resulting alarm fatigue among healthcare providers severely undermines patient safety and clinical responsiveness. The identified necessity for patient-specific alarm thresholds, capable of adapting to individual physiological baselines, clearly supports the need for intelligent, adaptive alarm systems as proposed by my research.

Secondly, multiple studies in section 2.2 have illustrated the substantial benefits of integrating multivariate physiological data for improving alarm specificity and accuracy. While these studies effectively demonstrate reductions in false positives through multi-parameter data integration, their approaches remain largely reactive, focusing on immediate physiological changes without predictive capacities. My research addresses this limitation by leveraging multivariate integration for predictive modeling, providing earlier and clinically meaningful warnings.

Thirdly, machine learning techniques have proven effective in enhancing alarm classification accuracy, as evidenced by section 2.3. These works highlight ML's capability to accurately distinguish true alarms from false positives, significantly reducing unnecessary alarms. Nevertheless, these studies predominantly emphasize real-time classification without exploring the potential of forecasting future clinical deteriorations. My research bridges this gap by applying machine learning not only to classify but also to predict critical deterioration events, transitioning from reactive to proactive patient monitoring.

Finally, section 2.4 establishes the feasibility and accuracy of forecasting patient conditions using machine learning, underscoring its capability for earlier, more effective clinical interventions. My research builds upon these results, applying similar techniques to a new problem: sustained hypotensive events.

My research directly addresses these critical gaps by reframing what an alarm should provide. From a clinical perspective, alarms should act as an early detection system, aiding nurses in their ability to prevent critical deterioration. Currently, this is achieved by sounding an alarm every time a patient could need intervention, which empirically causes too many false positives. This literature review supports my hypothesis that it is possible to accurately and precisely predict critical deterioration for hypotensive events by analyzing multi-source physiological data using machine learning techniques. This shift from reactive to proactive alarm systems will significantly mitigate alarm fatigue, enhance timely clinical interventions, and ultimately improve patient outcomes in critical care environments.

3 Methods

3.1 Data Collection

The data utilized for this research was sourced from a comprehensive retrospective database maintained by The Hospital for Sick Children (SickKids), which contains high-resolution waveform measurements recorded from critical care monitoring devices. The SickKids database was selected for its robust volume and diversity of physiological data, of which 19500 hours of waveform data across 1162 patients was used to enable the rigorous analyses necessary for developing predictive machine learning models for clinical deterioration.

Three primary waveforms were selected due to their clinical relevance and documented significance in predicting critical health events in intensive care contexts, as highlighted by Stevens et al. [7] and Hever et al. [8]. These waveforms included:

- **Arterial Line Blood Pressure (ABP):** This waveform was recorded continuously at 100 Hz. The ABP measurements were obtained through an arterial catheter, a thin, flexible tube inserted directly into a patient's artery, providing continuous, highly accurate, and invasive blood pressure readings. Continuous ABP monitoring is widely regarded as the gold standard in critically ill patients due to its precision and real-time responsiveness, allowing for the immediate detection of hypotensive episodes.

Figure 3.1.1: ABP mmHg waveform, 2s window

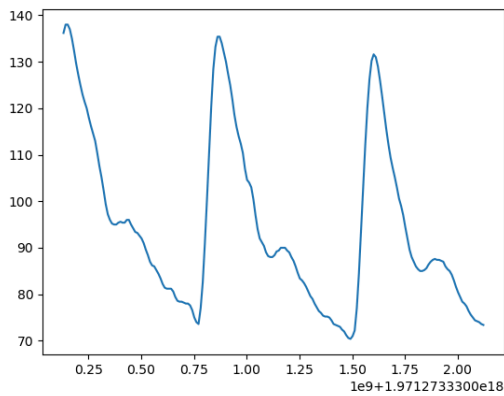
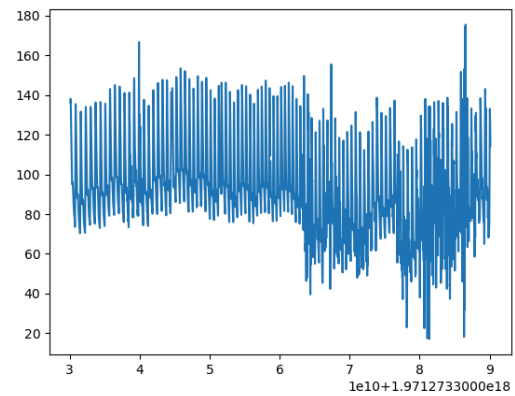


Figure 3.1.2: ABP mmHg waveform, 60s window



- Pulse Oximeter (SpO₂): This signal was collected at 100 Hz via a pulse oximetry device. Pulse oximetry is a non-invasive technique that measures peripheral oxygen saturation, an essential parameter reflecting the adequacy of oxygenation and pulmonary function. It is crucial in detecting hypoxemia and has been extensively utilized as an early indicator of respiratory or circulatory compromise.
- Electrocardiogram (ECG): ECG data was recorded at 200 Hz using standard bedside monitors. This waveform captures the electrical activity of the heart, facilitating real-time monitoring of heart rate, rhythm disturbances, and cardiac electrical anomalies. Given its non-invasive nature and diagnostic richness, ECG provides indispensable insights into cardiovascular health and is integral for predicting critical cardiac events such as arrhythmias.

The choice of these waveforms is justified by their established clinical utility and proven efficacy in previous studies to provide essential physiological context for predicting adverse health outcomes [11], [13]. High-frequency sampling of these waveforms was deliberately chosen to ensure the capturing of subtle physiological variations and transient events, crucial for accurate and timely predictive analytics. Furthermore, the multi-modal approach (integrating combinations of ABP, SpO₂, and ECG) aligns with evidence from existing literature [7], [9], employing the benefits of multivariate signal integration for improved predictive performance and clinical relevance.

In summary, the data collected for this research provides a comprehensive and clinically relevant foundation, crucial for the development and validation of a proactive, predictive model designed to enhance early detection and intervention for critical deteriorations in patient health.

3.2 Data Preprocessing

Data preprocessing consisted of several critical steps designed to ensure that the model input data accurately and reliably represents clinically relevant physiological events, specifically hypotensive episodes.

3.2.1 Event Labelling

Hypotensive events were meticulously labeled based on clinical definitions for systolic, mean, and diastolic blood pressures under anesthesia, as established in relevant literature [19]. Events were identified when arterial line blood pressure measurements continuously remained below the predefined threshold set at -2SD (as provided in Table 3.2.1, and Appendix A) for at least 30 consecutive seconds. This duration was chosen intentionally under clinical guidance to ensure clinical relevance by focusing the model's predictive capabilities on critical, sustained hypotensive events, which have greater clinical significance and require urgent medical intervention. The final labeled dataset comprised 5,842 independent hypotensive events. A new event was defined only after the blood pressure recovered above threshold levels for 30 consecutive seconds, distinguishing sequential events clearly and ensuring independence among labeled episodes.

Table 3.2.1: Systolic non-invasive blood pressure during surgical phase in relation to age [19]

Surgical Phase	Male					Female				
	Systolic non-invasive blood pressure (mm Hg)					Systolic non-invasive blood pressure (mm Hg)				
Age	-2SD	-1SD	0SD	+1SD	+2SD	-2SD	-1SD	0SD	+1SD	+2SD
0	31	41	54	72	103	31	39	50	63	83
1 month	40	51	63	79	104	40	50	62	75	95
2 months	44	55	66	81	104	44	54	66	80	99
3 months	47	57	69	83	105	47	57	69	82	102
4 months	49	59	71	85	106	49	59	71	85	104
5 months	50	61	72	86	106	50	61	73	87	106
6 months	52	62	73	87	107	52	63	74	88	107
9 months	55	65	76	89	108	55	66	78	91	110
1 year	57	67	78	91	109	57	68	80	93	112
2 years	62	72	83	96	113	62	73	85	98	116
3 years	65	75	86	99	117	65	75	87	100	118
4 years	66	77	89	102	120	66	77	89	102	120
5 years	68	78	90	104	123	67	78	90	104	122
6 years	68	79	91	106	124	68	79	91	105	124
8 years	69	81	93	107	127	69	80	93	108	127
10 years	70	82	94	109	128	70	81	94	109	129
12 years	71	83	95	111	130	71	82	94	109	129
14 years	72	84	97	112	133	72	83	95	109	129
16 years	74	86	99	114	135	73	84	96	110	130
18 years	76	88	101	117	137	75	86	97	111	131

3.2.2 Data Chunking (Windowing)

Physiological signals were segmented into uniform 30-second windows, consistent with the criteria used for event labeling. This choice of window length was explicitly linked to the clinical definition of hypotension, ensuring each data chunk would potentially capture a clinically significant event. Additionally, standardizing window lengths facilitates the comparability and consistent feature extraction necessary for robust machine learning model training. The SickKids database contained a total of 70,232,358 30-second windows, using a stride of 2 seconds. Using a stride of 2 seconds was necessary to capture and label all sustained hypotensive events, as the systolic, mean, and diastolic blood pressures needed for event labelling were calculated at 0.5Hz. For model training, a stride of 30 seconds was used to avoid repetitive samples, which resulted in approximately 2,341,100 30-second windows.

3.2.3 Outlier and Artifact Detection

Outlier detection was conducted based on established thresholds from clinical standards. Artifacts identified included physiologically implausible measurements such as diastolic blood pressure below 3 mmHg, systolic blood pressure equal to or exceeding 250 mmHg, systolic-diastolic pressure differentials less than or equal to 5 mmHg, etc. These artifacts were flagged for event labelling purposes; the measurement at the specific timestamp was replaced with NaN and did not bias a window towards either class; section 3.2.4 will cover how missing data was handled in greater detail. Although these artifact-containing segments were flagged, they were intentionally retained in the training dataset. This decision was justified by the rationale that an effective predictive model should inherently learn to distinguish between genuine physiological signals indicative of clinical events and artifacts that do not correlate with patient deterioration. This approach aligns with previous work, which suggests robustness of machine learning models improves when trained to recognize and implicitly filter out artifacts [13]. The artifact occurrences out of 70,232,358 windows are detailed in Table 3.2.2.

Table 3.2.2: Artifact Occurrences in SickKids Dataset out of 70,232,358 total windows analyzed

Criteria	# Windows	% Windows
Diastolic BP < 3 mmHg	2,336,710	3.327%
Systolic BP $\geq$ 250 mmHg	1,019,514	1.452%
Systolic BP — Diastolic BP $\leq$ 5 mmHg	4,631,180	6.594%
Diastolic — Mean > 0 mmHg	398,881	0.568%
Mean — Systolic > 0 mmHg	389,081	0.554%
Windows with at least one artifact	5,322,798	7.579%

3.2.4 Handling Missing Data

To analyze the missingness of data within the 30-second ABP windows used for event labelling (which entirely encompasses the windows that were used in model training), percent missing data thresholds were used, from 0% to 100% in 10% increments. Instead of directly eliminating/skipping windows with artifacts present, the timestamps of artifacts were replaced with NaN, and the number of NaN's in the window was counted. The event labelling procedure (detailed in section 3.2.1) was run for each of these thresholds, where windows with NaN percentages lower than or equal to the threshold were included. The results in Table 3.2.3, Figure 3.2.4, and Figure 3.2.5 illustrate how many windows fulfilled the missing data threshold for each percent threshold, and how many hypotensive events were captured from that subset of windows.

Under clinical guidance, we determined that eliminating all windows with any missing data or artifacts present (i.e. 0% missing data threshold) was an unreasonable condition due to it being unrealistic; events in the ICU will commonly cause at least some missing values. We determined that missing data could be considered clinically acceptable if up to 30% of the data within a single 30-second window (equivalent to 9 seconds) was unavailable (NaN or artifact). Windows excluding this threshold were excluded from event labelling and training datasets, resulting in 5842 hypotensive events labelled in 36,926,570 30-second windows.

Table 3.2.3: Number of windows and hypotensive events captured for each missing data threshold

Missing Data Threshold (i.e. % of NaN's in window $\leq$ threshold)	# of windows (stride = 2s)	# of additional windows	# of hypotensive events captured from windows	# of additional hypotensive events captured from windows
0%	20,796,718	---	2814	---
10%	23,632,279	2,835,561	3095	281
20%	29,120,621	5,488,342	4587	1492
30%	36,926,570	7,805,949	5842	1255
40%	61,504,222	24,577,652	7015	1173
50%	65,157,083	3,652,861	7226	211
60%	65,876,849	719,766	7467	241
70%	66,056,999	180,150	7553	86
80%	66,375,088	318,089	7802	249
90%	66,549,786	174,698	7932	130
100%	70,232,358	3,682,572	8232	300

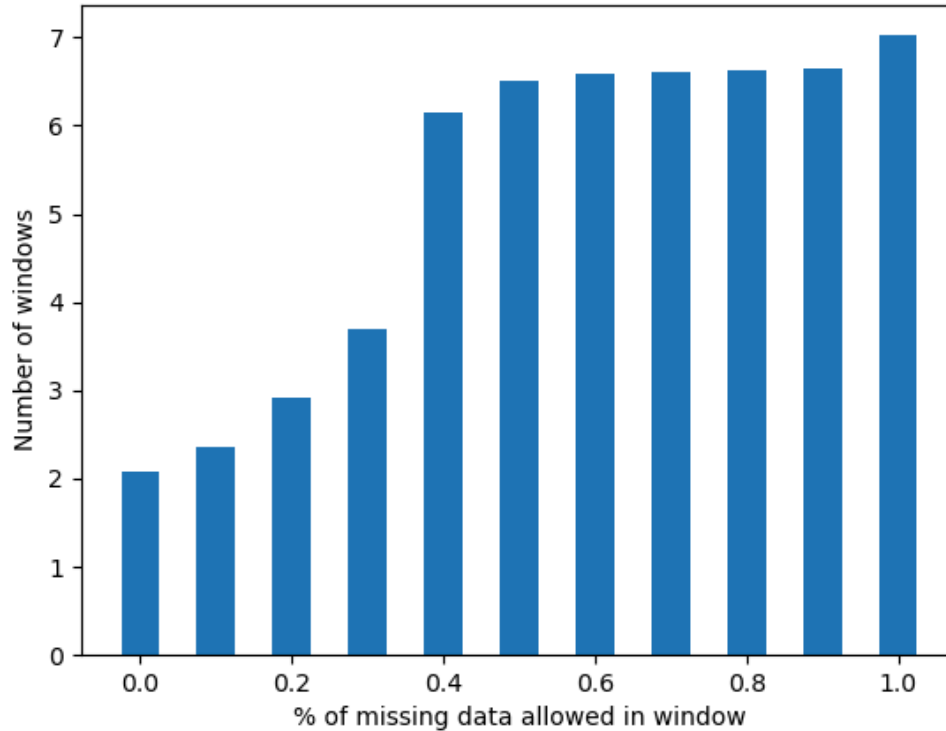
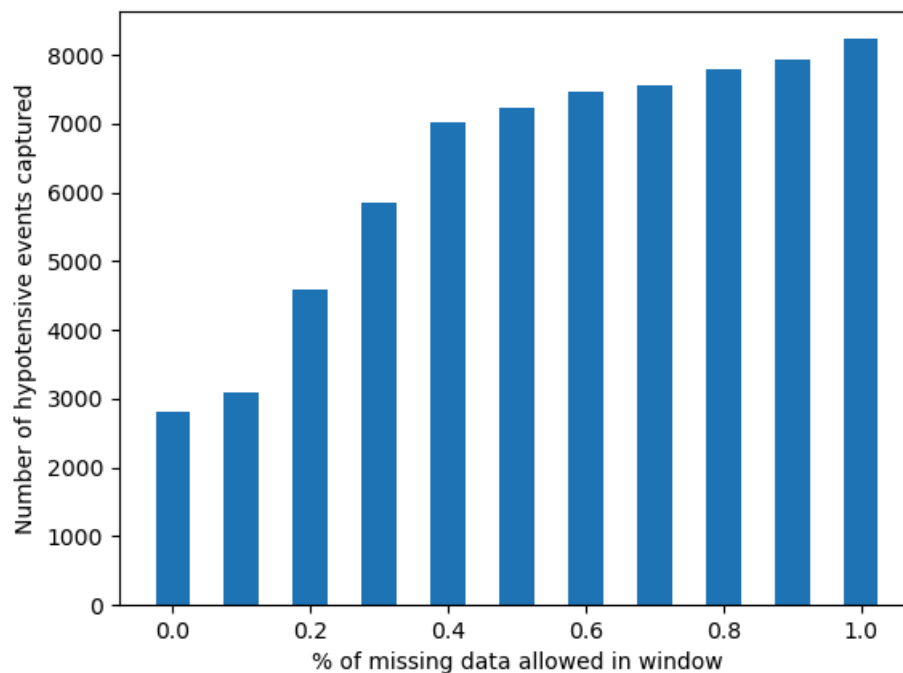
Figure 3.2.4: Number of 30-second windows (x 1e7) vs % missing data allowed in window

Figure 3.2.5: Number of hypotensive events labelled vs % missing data allowed in window



3.2.5 Positive/Negative Class Imbalance

Given the substantial class imbalance due to the relatively small number of positive examples (5842 hypotensive event starts $\times$ 10 30-second windows in a 5 minute prediction horizon; explained further in section 3.3) compared to negative examples (all remaining $\sim$ 1,230,900 30-second windows with 30s stride), a balanced training strategy was adopted. Specifically, random sampling from the negative class was employed in each training epoch to maintain a 1:1 ratio of positive to negative examples. This technique ensures the model is effectively trained to identify true positives, reducing bias toward the majority negative class, a common and well-documented concern in predictive modeling within healthcare datasets [16].

In summary, the detailed preprocessing pipeline was carefully designed and justified to ensure that the data input into the predictive model accurately represents clinically significant events while maintaining the integrity, robustness, and generalizability required for high-performing predictive analytics in critical care scenarios.

3.3 Model Creation & Architecture

The specific goal of this research is to develop a machine learning model capable of predicting whether a patient will experience a hypotensive episode within a 5-minute prediction horizon, based on a 30-second window of preceding physiological waveform data. In other words: given the past 30 seconds of waveform data for a patient, will the patient experience a period of sustained hypotension within 5 minutes? This was formulated as a binary classification problem, where the desired model output is:

- **1** if a hypotensive event occurs within the subsequent five minutes
- **0** if no such event occurs within the prediction window

This structure was designed to align with the research goal of proactive, rather than reactive, monitoring—allowing for early intervention prior to the onset of sustained hypotension.

3.3.1 Model Inputs

Each model was trained using 30-second segments of high-frequency physiological data, comprising one or more of the previously introduced waveforms:

- Arterial Line Blood Pressure (ABP) at 100 Hz
- Pulse Oximeter (SpO₂) at 100 Hz
- Electrocardiogram (ECG), downsampled to 100 Hz for consistency

These modalities were chosen due to their known relevance in monitoring cardiovascular and respiratory function, as discussed in [7], [8], and [13]. Their combined use supports multivariate signal fusion and enables the model to learn more complex physiological interactions.

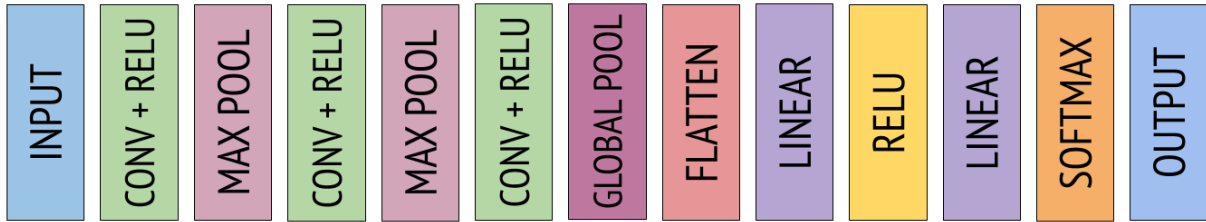
As mentioned in section 3.2.5, there are 5842×10 positive inputs, since there are 5842 hypotensive event start timestamps, and using a 5-minute prediction horizon includes ten 30-second windows that will be included in the positive class. There are approximately 1,230,900 30-second windows using a 30s stride, which make up the negative class. To reiterate, random sampling will be used from the negative class to avoid issues with training with a large class imbalance [16].

3.3.2 CNN-Based Models

Two models were prototyped using the convolutional neural network (CNN) architecture. CNN's were explored due to their demonstrated effectiveness in processing raw time-series data and extracting local temporal features. Additional benefits include their efficiency in parallelizing computations and their ability to capture local temporal dependencies, which is ideal for short, fixed-length segments such as the 30-second windows used in this study. These characteristics have been validated in prior physiological signal processing literature [10].

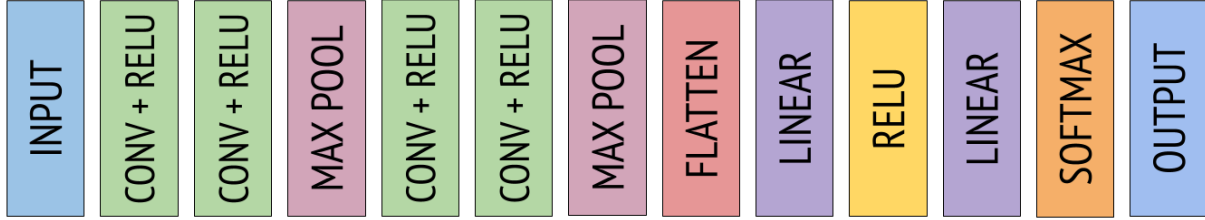
The first CNN model adopted a modular design, using a three-layer convolutional block followed by an adaptive global max pooling layer to collapse temporal dimensions into a fixed feature representation. This was then passed through a two-layer fully connected classifier. The use of global pooling helped reduce sensitivity to input length variation and allowed for more robust feature aggregation.

Figure 3.3.1: 3-Layer CNN Model Design



The second CNN model consisted of four 1D convolutional layers with progressively increasing feature map dimensions, followed by two fully connected layers. Max pooling layers were interspersed after the second and fourth convolutional layers. ReLU activations were applied after each layer to introduce non-linearity. This architecture was optimized to extract deep temporal features over multiple scales.

Figure 3.3.2: 4-Layer CNN Model Design



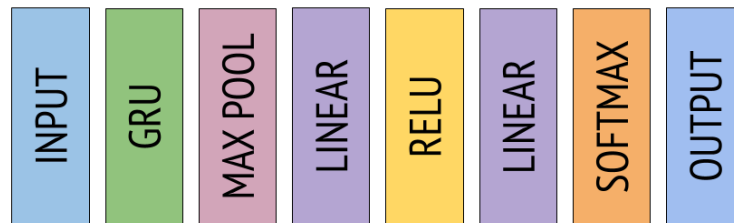
The code for the model definitions can be found in Appendix B.

3.3.3 RNN-Based Models

Recurrent models, especially GRUs, were chosen for their proven capability to model temporal patterns in physiological signals and their ability to ‘forget’ irrelevant information (such as missing values, ideally). Their application in ICU alarm modeling in [8] and [15] has shown promise in identifying subtle precursors to deterioration that CNNs may miss. To model sequential dependencies and longer-range temporal relationships, two recurrent neural network architectures were developed using Gated Recurrent Units:

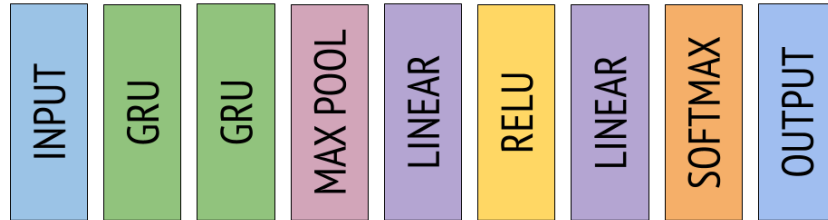
The first RNN model used a single GRU layer with 64 hidden units. After processing the sequence, a global max pooling layer aggregated the temporal outputs into a single feature vector. This was followed by two ReLU-activated fully connected layers. The input dimension matches the number of waveforms used as input, assuming a feature vector per time step.

Figure 3.3.3: 1-Layer RNN Model Design



The second RNN model included a stacked GRU architecture with two layers and the same hidden size. This deeper model was designed to capture more complex temporal interactions across multiple physiological streams.

Figure 3.3.4: 2-Layer RNN Model Design



The code for the model definitions can be found in Appendix B.

All models employed ReLU activations to promote sparse representations and mitigate vanishing gradient issues. The architectures were designed to be lightweight enough for real-time inference while maintaining sufficient depth to extract clinically meaningful features.

In summary, this dual-model exploration (CNN vs. RNN) was undertaken to compare the relative strengths of spatial vs. sequential feature extraction for the prediction of hypotensive events. Both architectures were designed to process raw, high-frequency physiological waveforms and output clinically actionable predictions within a time frame that supports proactive intervention in intensive care settings.

3.4 Model Training

To evaluate the predictive performance of the models, the dataset was randomly split into training, validation, and testing subsets in an 80:10:10 ratio. The resulting partitions included 90,570 training samples, 11,980 validation samples, and 11,979 test samples. This split was applied while maintaining an approximately equal distribution of positive (hypotensive) and negative (non-hypotensive) samples across each subset, ensuring class balance was preserved by undersampling the majority class (non-hypotensive windows) for each epoch. As discussed in the preprocessing section, this strategy mitigates the risk of bias toward the majority class and aligns with best practices for handling imbalanced datasets in binary classification tasks [16].

Model training was conducted using the Adam optimizer with an initial learning rate of 0.01, chosen for its adaptive learning capabilities and computational efficiency, especially for problems with sparse gradients. Training was performed for 10 epochs. At the end of each epoch, validation accuracy and loss were recorded, and the model with the highest validation accuracy (or lowest validation loss in case of ties) was retained for final evaluation on the test set. This approach guards against overfitting and ensures that model selection is based on generalizable performance rather than training-specific optimization.

The binary cross-entropy loss function was employed as the training criterion. As the target is a binary classification outcome (presence or absence of a hypotensive event), this loss function was appropriate for penalizing misclassifications and calibrating probabilistic outputs.

Two data-cleaning strategies for handling missing values were explored:

- **Mean Imputation and Standardization:** In this approach, missing values (NaNs) within each 30-second window were replaced with the mean of the non-missing values in the same window. Following this, the entire segment was standardized by subtracting the mean and dividing by the standard deviation. This technique assumes that the local statistical characteristics of

a signal segment are representative of the underlying physiological process and that normalization improves model convergence. It is particularly effective when signal amplitude or scale variations can obscure underlying temporal features.

- **Out-of-Domain Replacement:** As an alternative, missing values were replaced with a constant value of -1, an out-of-domain input for all waveform modalities. This strategy enables the model to learn that the presence of this specific value indicates missingness. Such a method has been used effectively in previous ICU-focused studies [8], where it was shown that models can implicitly learn to ignore or discount the impact of missing values when they are clearly demarcated.

Both approaches are valid and carry different assumptions about how models handle incomplete data. Mean imputation with standardization maintains signal continuity and normalizes variance, potentially benefiting models that rely on subtle feature scaling, but potentially removes important information (the mean and variance) from the inputs. Out-of-domain encoding of missing data, meanwhile, offers a more direct and interpretable method for exposing models to the patterns of data absence, which can itself be informative in ICU contexts.

Summarizing, model training was carefully structured to ensure reproducibility, generalization, and clinical relevance. The combination of stratified data partitioning, class balancing, robust optimizer settings, and principled handling of missing data supports the validity and applicability of the predictive modeling approach in real-world critical care settings.

4 Model Performance Results

This chapter presents and highlights the key results of the model development process, focusing on the predictive accuracy and classification metrics across different architectures and data configurations. Table 4.1 contains the training results—the best training accuracy, training loss, validation accuracy, validation loss the model was able to achieve from the epoch with highest validation accuracy. Model selection was guided by early stopping based on validation set performance—specifically, the model from the epoch with the highest validation accuracy (or lowest validation loss) was selected for final evaluation on the test dataset. Full training loss, training accuracy, validation loss, and validation accuracy curves can be found in Appendix C.

Table 4.1: Model training configurations and performances

Model	Input Waveforms	Data Cleaning	Train Acc	Train Loss	Val Acc	Val Loss	Test Acc	Test Loss
3-Layer CNN	Arterial Line	Standardize	51.73%	0.6925	51.87%	0.6925	51.80%	0.6925
	Pulse Oximeter	Standardize	66.82%	0.604	64.55%	0.6234	65.30%	0.6154
4-Layer CNN	Arterial Line	Standardize	51.60%	0.6931	52.05%	0.6931	51.80%	0.6931
1-Layer RNN	Arterial Line	Standardize	69.15%	0.5809	70.41%	0.5844	70.20%	0.5833
	Arterial Line	OFF	71.32%	0.5537	72.67%	0.5689	71.78%	0.5747
	Pulse Oximeter	OFF	71.03%	0.5703	70.87%	0.5798	69.33%	0.5982
2-Layer RNN	Arterial Line	OFF	72.59%	0.5467	78.56%	0.4910	72.65%	0.5610
	Pulse Ox. + ECG	OFF	65.31%	0.6026	62.64%	0.6295	63.32%	0.6116
	Art. Line + Pulse Ox.	OFF	90.23%	0.2575	91.12%	0.3632	91.79%	0.3939
	Art. Line + Pulse Ox.+ ECG	OFF	92.35%	0.2689	92.53%	0.3595	92.10%	0.3507

During experimentation, two primary types of architectures were evaluated: convolutional neural networks (CNNs) and recurrent neural networks (RNNs). The upper rows of the training results in Table 4.1 correspond to the CNN-based models. While these 1D CNN models were somewhat effective at extracting local temporal features, they consistently underperformed compared to RNN-based models, particularly in terms of classification accuracy and recall on the validation and test datasets. Due to limited computational resources and the aforementioned underperformance of CNN-based models, following experiments were prioritized toward RNN models, especially those incorporating multiple physiological inputs.

The most successful models were the two-layer gated recurrent unit (GRU) networks. These RNNs were trained with combinations of arterial line blood pressure, pulse oximeter and ECG data, with the most successful multimodal combinations being Art Line + Pulse Ox. and Art Line + Pulse Ox. + ECG. Notably, the best results were obtained using combinations of raw (non-cleaned) waveform inputs, suggesting that models were capable of internally learning to distinguish signal from noise and handling missing or anomalous values without explicit preprocessing.

Table 4.2: Confusion Matrices for test split results for top 2 performing models

2-Layer RNN: Arterial Line + Pulse Oximeter			2-Layer RNN: Art. Line + Pulse Ox. + ECG		
	Actual Positive	Actual Negative		Actual Positive	Actual Negative
Predicted Positive	5813	834	Predicted Positive	5820	803
Predicted Negative	150	5182	Predicted Negative	143	5213
Accuracy	91.79%	F1 Score:	Accuracy	92.10%	F1 Score:
Precision	87.45%	0.922	Precision	87.88%	0.9248
Recall	97.48%		Recall	97.60%	

The best-performing RNN models achieved a test accuracy of approximately 92%, indicating strong generalization to unseen data. The associated confusion matrices in Table 4.2 reveal a slightly imbalanced performance profile: both models exhibited a higher number of false positives than false negatives, with ~87% precision and ~97% recall for both, and an F1 score of ~0.92.

This skew leads to a higher recall than precision, which, while not perfectly aligned with the research objective of reducing false positives, remains clinically acceptable. From a critical care perspective, the cost of a false negative—failing to warn of an impending hypotensive event—is far more detrimental than a false positive, which may lead to a momentary increase in clinical workload but poses no direct risk to patient safety. The high recall thus reflects the model's effectiveness in capturing the majority of true deterioration events, supporting the broader clinical goal of providing early, actionable warnings.

5 Discussion of Results

The results obtained from the experiments directly support the central research objective of this thesis: to develop a predictive machine learning model capable of reducing false alarms and providing earlier, actionable warnings for critical hypotensive events in intensive care units. This discussion interprets the findings within the broader clinical and technical context, offering insight into model performance, architecture selection, and the practical implications for intelligent alarm systems.

5.1 RNNs Outperformed CNNs

One of the most prominent findings was the superior performance of recurrent neural networks (RNNs), specifically those using Gated Recurrent Units (GRUs), over convolutional neural networks (CNNs). This is likely attributed to the RNNs' ability to model long-range temporal dependencies, which is essential when working with time-series physiological data. High-resolution signals like arterial line blood pressure and pulse oximetry carry temporal dynamics that unfold over several seconds; RNNs can maintain state across time steps and capture these evolving patterns. CNNs, on the other hand, are inherently better suited to extracting local, spatial features, and may have struggled to capture dependencies that span beyond their receptive fields.

However, it is important to note that this result may not be universally generalizable. The observed underperformance of CNN models could partially stem from implementation limitations, such as suboptimal architectural depth, kernel size selection, or pooling strategies. Further experimentation with more sophisticated temporal convolutional networks or hybrid architectures could potentially mitigate these limitations and narrow the performance gap.

5.2 Raw Data Performed Well

Another notable observation is that the models trained on raw waveform data outperformed those trained on preprocessed (standardized) data. This suggests two

key points. First, the GRU-based architectures demonstrated a strong capacity to internally manage artifacts and missing values without explicit preprocessing, possibly learning to identify and ignore spurious signals or noise. Second, standardizing the data—especially without including the mean and standard deviation as additional input features—may inadvertently remove vital contextual information. In waveform data, the absolute values and baseline trends often carry clinically significant meaning. By normalizing these away, the model may lose access to precisely the information needed to identify early warning signs of physiological deterioration.

This result sheds light upon a broader point about machine learning in healthcare: data cleaning, while important, must be carefully balanced against the risk of erasing signal-rich variability. The finding also suggests that in certain contexts, minimal preprocessing may allow the model to retain more raw physiological nuance.

5.3 Best Configuration: Two-Layer GRU with Multi-Source Input

The most consistent, accurate, and precise performances were achieved by two-layer GRU models trained on multi-source physiological data—specifically, arterial line blood pressure and pulse oximeter signals, with or without ECG. The additional GRU layer likely enabled the model to capture more abstract temporal relationships, expanding its capacity to recognize patterns that single-layer models may miss. Stacked RNNs are known to enhance representational power without overly complicating training, particularly when combined with max pooling layers to mitigate the effects of vanishing gradients.

Additionally, the inclusion of multiple physiological signals significantly improved predictive performance. This result aligns with findings from [7] and [8], which demonstrated that integrating multivariate inputs allows models to capture complex physiological interactions that would be missed when relying on a single modality. For instance, patterns in pulse oximetry and ECG may precede or coincide with changes in blood pressure, providing early signals of impending sustained

hypotension. By training on combined data sources, the model effectively simulates the integrative reasoning a human clinician might use, thereby improving predictive sensitivity and robustness.

5.4 Recall-Precision Tradeoff

The two best-performing models exhibited a high recall of 97% and a relatively high precision of 87%. Although one of the original goals of this research was to reduce false positives and combat alarm fatigue, the results demonstrate a necessary tradeoff in favor of clinical safety. In the context of critical care, false negatives—missed warnings of hypotensive deterioration—pose a greater risk to patient health than false positives. The model’s high recall means that nearly all true hypotensive events were identified, minimizing the chances of a missed deterioration.

At the same time, the precision of 87% is significantly higher than that of current threshold-based alarm systems, where studies report false alarm rates as high as 85–99% [3]. This indicates that while some false positives remain, the model substantially improves alarm precision and potentially specificity, reducing the volume of unnecessary alerts while preserving sensitivity to critical events. Thus, this research advances both goals: reducing false alarms and ensuring early, actionable warnings.

5.5 Summary and Implications of Results

This work successfully demonstrates that a machine learning model can be trained to accurately and precisely predict critical deterioration for hypotensive events within a five-minute horizon. By leveraging multi-source physiological data and using a two-layer GRU architecture, the system provides earlier alerts than conventional reactive threshold-based alarms. This approach not only reduces alarm fatigue but also enhances clinical decision-making by shifting the paradigm from reaction to prediction.

The findings reinforce the potential for intelligent, personalized monitoring systems to transform alarm management in intensive care. By providing adaptive, context-aware, and forward-looking alerts, such systems align more closely with how clinicians think and act. In doing so, this research moves the field closer to a future of proactive, precision-based care in high-stakes clinical environments.

6 Limitations

One important limitation lies in the method used to split the dataset into training, validation, and test sets. The data was randomly partitioned without accounting for patient identity, meaning that different hypotensive events from the same patient could appear in multiple subsets. This overlap introduces the possibility of data leakage, where patterns unique to an individual’s physiology may be learned during training and then reappear in the validation or test phase, potentially inflating performance metrics. While this may not entirely invalidate the results—especially given the model’s performance across varied input configurations—it does mean that generalization to completely unseen patients should be further validated with a patient-level split. Ensuring that data from any given patient appears in only one subset would provide a more rigorous assessment of the model’s ability to generalize across diverse patient populations.

Additionally, although raw waveform data led to strong results, relying solely on deep learning with minimal preprocessing may limit interpretability and overlooks potentially valuable domain knowledge embedded in engineered features. Moreover, the reliance on only three physiological inputs—ABP, SpO₂, and ECG—excludes other contextual and clinical variables that could provide complementary information about the patient’s condition.

7 Future Work

Several directions could be pursued to further improve and extend this research:

7.1 Feature Engineering and Signal Enrichment

Future iterations of this work could benefit from incorporating handcrafted features derived from the waveform data. For example, frequency-domain representations using Fourier transforms or wavelet decompositions could help capture underlying signal patterns not immediately obvious in the time domain. Similarly, physiological event markers such as heartbeat intervals or variability indices (e.g., heart rate variability) may offer additional insights for predicting deterioration.

7.2 Refined Event Labelling Criteria

The current hypotensive event labeling relies on a static threshold being crossed for a continuous duration. Future work could explore more dynamic labeling schemes, such as requiring blood pressure to return above a certain statistical threshold (e.g., -1 SD) before an event is considered resolved. These approaches may improve event specificity and better mimic the nonlinear and evolving nature of real-world clinical deterioration.

7.3 Incorporation of Contextual and Demographic Features

Integrating patient metadata such as age, sex, height, weight, surgery status, and clinical context (e.g., operating room vs. ICU) into the model architecture could significantly enhance predictive accuracy and personalization. Contextual information can help differentiate between patients who may present similarly in waveform data but have very different risk profiles due to underlying conditions or interventions.

7.4 Exploring Alternative Architectures

While GRU-based RNNs provided strong baseline results, alternative deep learning models could offer performance gains. Long Short-Term Memory (LSTM)

networks, for example, might be better suited for learning long-range dependencies due to their more complex memory gates. Transformer-based architectures—currently state-of-the-art in many sequential domains—could also be explored. Although typically more resource-intensive, transformers may offer better parallelism and capture both short- and long-term dependencies in the waveform data.

7.5 Real-Time Implementation and Clinical Validation

A key next step would be the deployment of the trained model in a real-time monitoring setting. This would allow evaluation of its performance under operational constraints and in the presence of noisy, streaming data. Collaborations with clinicians could also help assess the interpretability, trustworthiness, and usability of the model in practice.

7.6 Longitudinal Generalization

To test model robustness, future studies should employ a strict patient-level split and consider evaluating performance across different hospital units or institutions. This would ensure that the model generalizes beyond the specific characteristics of the SickKids dataset and holds potential for wider clinical application.

8 Conclusion

This thesis set out to address a critical challenge in intensive care: the high volume of non-actionable alarms generated by threshold-based monitoring systems, which contribute to alarm fatigue and compromise patient safety. By shifting the paradigm from reactive to predictive alarm systems, this work introduced a machine learning-based approach to forecast hypotensive events within a five-minute horizon using multi-source physiological waveform data.

The project successfully demonstrated that an intelligent alarm system, driven by machine learning and grounded in multivariate time-series data, can significantly improve the accuracy and timeliness of hypotensive event detection. Specifically, two-layer GRU models trained on arterial line blood pressure and pulse oximeter signals—optionally augmented with ECG—outperformed convolutional architectures and other baseline configurations. These models achieved high test accuracies (~92%), with exceptional recall (~97%) and robust precision (~87%), indicating their capacity to detect nearly all clinically significant hypotensive episodes while substantially reducing false positives compared to traditional systems.

Several key design decisions were validated through empirical results. The use of raw waveform data proved superior to standardized inputs, highlighting the importance of preserving physiological context in model training. Multivariate integration was also shown to enhance performance, as it enabled the model to capture the interdependent dynamics between cardiovascular and respiratory systems.

While the results are promising, the work acknowledges limitations such as the use of a randomly partitioned dataset that may have included data from the same patient across splits. Future studies should explore patient-level splits, incorporate contextual clinical data, and experiment with advanced model architectures like transformers to enhance generalizability and interpretability.

Ultimately, this research contributes a validated, data-driven framework for predictive alarm generation in critical care. By anticipating deterioration before it

becomes severe, the proposed system offers a more proactive and precise method of clinical intervention. This advancement not only reduces alarm fatigue but also supports earlier treatment, potentially improving patient outcomes and enhancing the overall quality of care in high-stakes environments.

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Appendix A

Table A.1: Systolic non-invasive blood pressure during surgical phase in relation to age [19]

Surgical Phase	Male					Female				
	Systolic non-invasive blood pressure (mm Hg)					Systolic non-invasive blood pressure (mm Hg)				
Age	-2SD	-1SD	0SD	+1SD	+2SD	-2SD	-1SD	0SD	+1SD	+2SD
0	31	41	54	72	103	31	39	50	63	83
1 month	40	51	63	79	104	40	50	62	75	95
2 months	44	55	66	81	104	44	54	66	80	99
3 months	47	57	69	83	105	47	57	69	82	102
4 months	49	59	71	85	106	49	59	71	85	104
5 months	50	61	72	86	106	50	61	73	87	106
6 months	52	62	73	87	107	52	63	74	88	107
9 months	55	65	76	89	108	55	66	78	91	110
1 year	57	67	78	91	109	57	68	80	93	112
2 years	62	72	83	96	113	62	73	85	98	116
3 years	65	75	86	99	117	65	75	87	100	118
4 years	66	77	89	102	120	66	77	89	102	120
5 years	68	78	90	104	123	67	78	90	104	122
6 years	68	79	91	106	124	68	79	91	105	124
8 years	69	81	93	107	127	69	80	93	108	127
10 years	70	82	94	109	128	70	81	94	109	129
12 years	71	83	95	111	130	71	82	94	109	129
14 years	72	84	97	112	133	72	83	95	109	129
16 years	74	86	99	114	135	73	84	96	110	130
18 years	76	88	101	117	137	75	86	97	111	131

Table A.2: Mean non-invasive blood pressure during surgical phase in relation to age [19]

Surgical Phase	Male					Female				
	Mean non-invasive blood pressure (mm Hg)					Mean non-invasive blood pressure (mm Hg)				
Age	-2SD	-1SD	0SD	+1SD	+2SD	-2SD	-1SD	0SD	+1SD	+2SD
0	21	28	37	49	71	21	27	34	45	62
1 month	26	33	41	51	70	26	32	40	51	67
2 months	27	34	42	53	70	27	34	42	53	69
3 months	29	35	43	53	70	29	36	44	54	71
4 months	30	36	44	54	70	30	37	45	55	72
5 months	31	37	45	55	70	31	38	46	56	73
6 months	31	38	45	55	70	31	39	47	57	73
9 months	33	39	47	56	71	33	40	49	59	75
1 year	34	40	48	57	72	34	42	50	60	76
2 years	36	43	51	61	75	37	44	52	63	78
3 years	38	45	53	63	78	38	45	54	64	79
4 years	38	46	54	65	80	39	46	55	66	81
5 years	39	46	55	67	82	39	47	56	67	82
6 years	39	47	56	68	83	40	48	57	68	84
8 years	40	48	58	70	86	41	49	59	71	86
10 years	42	50	60	72	88	42	50	60	72	88
12 years	43	51	61	74	89	43	51	61	74	89
14 years	44	52	63	76	91	44	52	62	75	90
16 years	45	54	64	77	92	45	53	63	76	92
18 years	47	55	66	79	94	47	55	65	78	95

Table A.3: Diastolic non-invasive blood pressure during surgical phase in relation to age [19]

Surgical Phase	Male					Female				
	Diastole non-invasive blood pressure (mm Hg)					Diastole non-invasive blood pressure (mm Hg)				
Age	-2SD	-1SD	0SD	+1SD	+2SD	-2SD	-1SD	0SD	+1SD	+2SD
0	15	20	27	38	58	15	19	25	35	53
1 month	17	23	30	39	56	18	23	29	38	56
2 months	18	24	31	40	55	19	24	31	40	57
3 months	19	25	31	40	55	19	25	32	41	57
4 months	19	25	32	41	55	20	26	32	42	58
5 months	20	26	32	41	55	20	26	33	42	58
6 months	20	26	33	41	55	21	27	34	43	59
9 months	21	27	34	42	56	22	28	35	44	59
1 year	21	28	34	43	56	22	29	36	45	60
2 years	22	29	36	45	58	23	30	38	47	61
3 years	23	30	38	47	61	24	31	39	48	62
4 years	23	31	39	49	63	24	32	40	49	64
5 years	23	31	40	50	64	24	32	40	50	65
6 years	23	32	41	51	66	24	33	41	51	66
8 years	24	33	42	53	68	25	34	43	53	68
10 years	25	34	43	54	69	26	35	44	54	69
12 years	26	35	45	55	70	27	36	45	55	70
14 years	28	37	46	56	71	29	37	46	57	72
16 years	30	38	47	57	72	31	39	48	59	74
18 years	32	40	49	59	73	33	41	50	61	78

Appendix B

Code B.1: 3-Layer CNN-Based Model

```
class CNNClassifierv1(nn.Module):
    def __init__(self):
        super(CNNClassifierv1, self).__init__()
        self.net = nn.Sequential(
            # Input shape: (batch_size, 1, 3000)
            nn.Conv1d(in_channels=1, out_channels=8, kernel_size=7, stride=2,
padding=3),
            nn.ReLU(),
            nn.MaxPool1d(kernel_size=2, stride=2),
            # Shape is now (batch_size, 8, ~750)

            nn.Conv1d(in_channels=8, out_channels=16, kernel_size=5, stride=1,
padding=2),
            nn.ReLU(),
            nn.MaxPool1d(kernel_size=2, stride=2),
            # Shape is now (batch_size, 16, ~375)

            nn.Conv1d(in_channels=16, out_channels=32, kernel_size=5,
stride=1, padding=2),
            nn.ReLU(),
            nn.AdaptiveMaxPool1d(1) # Global pooling to shape (batch_size,
32, 1)
        )

        self.classifier = nn.Sequential(
            nn.Flatten(),          # => (batch_size, 32)
            nn.Linear(32, 16),
            nn.ReLU(),
            nn.Linear(16, 1)      # => 1 logit
        )

    def forward(self, x):
        # x shape: (batch_size, 3000)
        x = x.unsqueeze(1)        # => (batch_size, 1, 3000)
        features = self.net(x)
        logits = self.classifier(features).squeeze(-1)
        return logits
```

Code B.2: 4-Layer CNN-Based Model

```
class CNNClassifierv2(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv_1 = nn.Conv1d(in_channels=1, out_channels=16,
kernel_size=15, stride=5, padding=0)
```

```

        self.conv_2 = nn.Conv1d(in_channels=16, out_channels=32,
kernel_size=7, stride=4, padding=0)
        self.conv_3 = nn.Conv1d(in_channels=32, out_channels=64,
kernel_size=5, stride=2, padding=0)
        self.conv_4 = nn.Conv1d(in_channels=64, out_channels=64,
kernel_size=5, stride=2, padding=0)

        self.fc_1 = nn.Linear(in_features=192, out_features=64)
        self.fc_2 = nn.Linear(in_features=64, out_features=1)

        self.maxpool1 = nn.MaxPool1d(kernel_size=3, stride=3)
        self.maxpool2 = nn.MaxPool1d(kernel_size=3, stride=3)
        self.relu = nn.ReLU()

    def forward(self, x):
        # run the model
        x = x.unsqueeze(1)
        x = self.relu(self.conv_1(x))
        x = self.relu(self.conv_2(x))
        x = self.maxpool1(x)

        x = self.relu(self.conv_3(x))
        x = self.relu(self.conv_4(x))
        x = self.maxpool2(x)

        x = x.flatten(start_dim=1)
        x = self.relu(self.fc_1(x))
        x = self.relu(self.fc_2(x))

        return x.squeeze(-1)

```

Code B.3: 1-Layer RNN-Based Model

```

class RNNClassifierv1(nn.Module):
    def __init__(self):
        super().__init__()
        self.rnn_1 = nn.GRU(input_size=1, hidden_size=64, num_layers=1,
batch_first=True)
        self.global_max_pool = nn.AdaptiveMaxPool1d(output_size=1)

        self.fc_1 = nn.Linear(in_features=64, out_features=64, bias=True)
        self.fc_2 = nn.Linear(in_features=64, out_features=1, bias=True)

        self.relu = nn.ReLU()

    def forward(self, x):
        # deal with potential nans

        # run the model

```

```

x = x.unsqueeze(2)
x = self.rnn_1(x)[0]
x = torch.squeeze(self.global_max_pool(torch.transpose(x, 1, 2)))

# x = x.flatten(start_dim=1)
x = self.relu(self.fc_1(x))
x = self.fc_2(x).squeeze(-1)

# x = self.softmax(x)

return x

```

Code B.4: 2-Layer RNN-Based Model

```

class RNNClassifierV2(nn.Module):
    def __init__(self):
        super().__init__()
        self.rnn_1 = nn.GRU(input_size=1, hidden_size=64, num_layers=2,
batch_first=True)
        self.global_max_pool = nn.AdaptiveMaxPool1d(output_size=1)

        self.fc_1 = nn.Linear(in_features=64, out_features=64, bias=True)
        self.fc_2 = nn.Linear(in_features=64, out_features=1, bias=True)

        self.relu = nn.ReLU()

    def forward(self, x):
        # deal with potential nans

        # run the model
        x = x.unsqueeze(2)
        x = self.rnn_1(x)[0]
        x = torch.squeeze(self.global_max_pool(torch.transpose(x, 1, 2)))

        # x = x.flatten(start_dim=1)
        x = self.relu(self.fc_1(x))
        x = self.fc_2(x).squeeze(-1)

        # x = self.softmax(x)

        return x

```

Appendix C

Figure C.1: Training Loss vs. Epoch

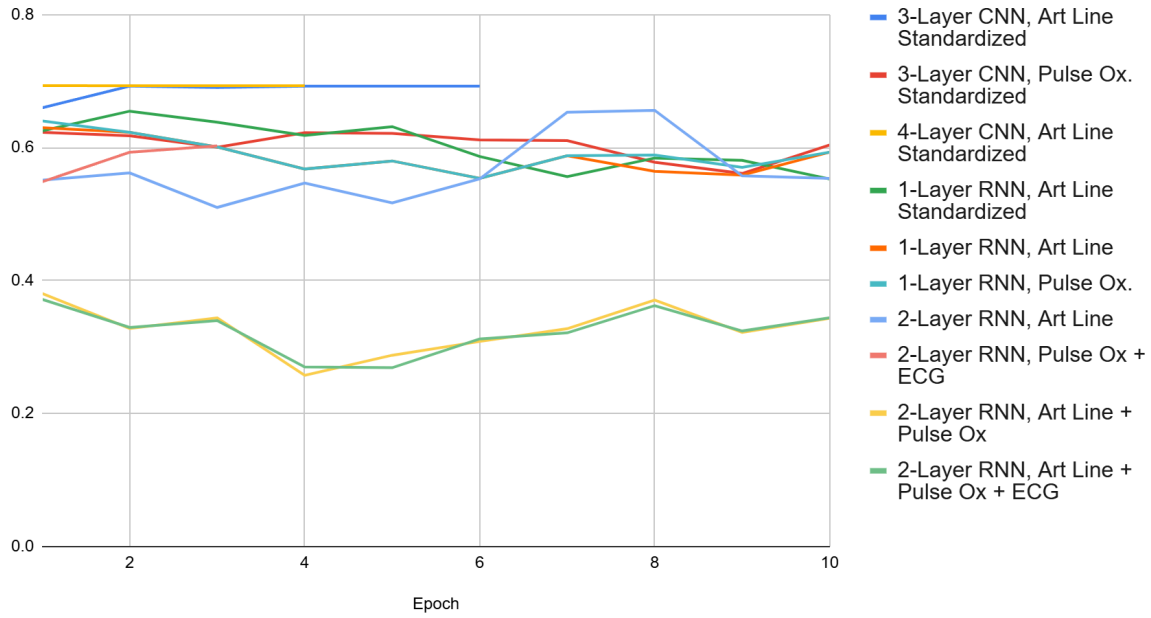


Figure C.2: Training Accuracy vs. Epoch

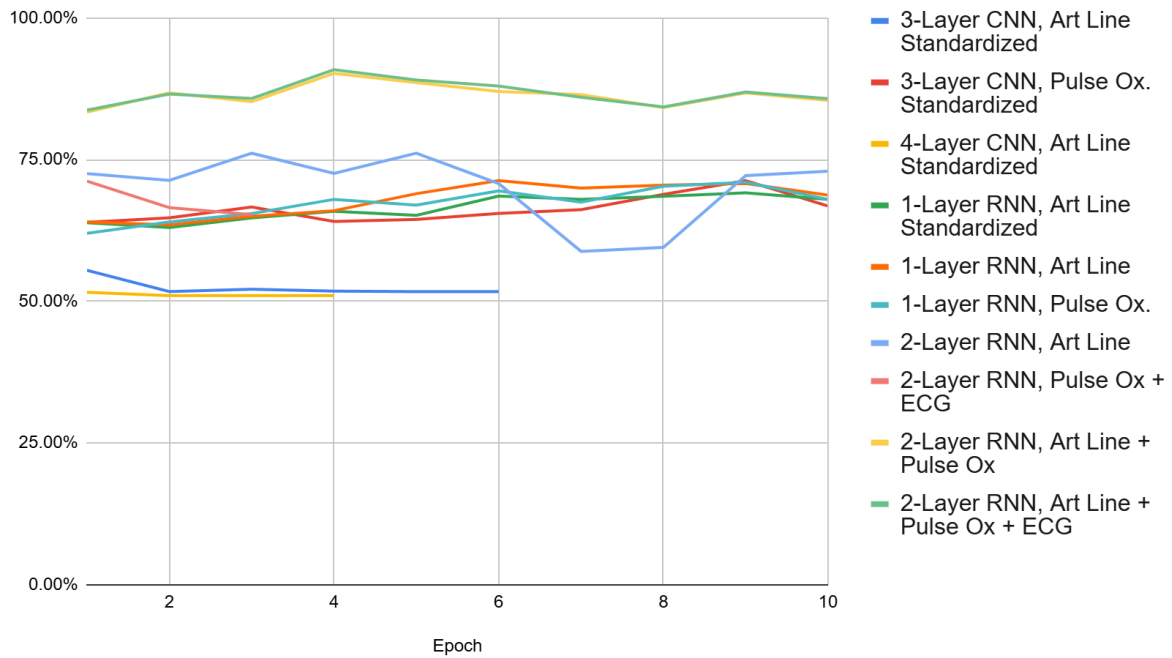


Figure C.3: Validation Loss vs. Epoch

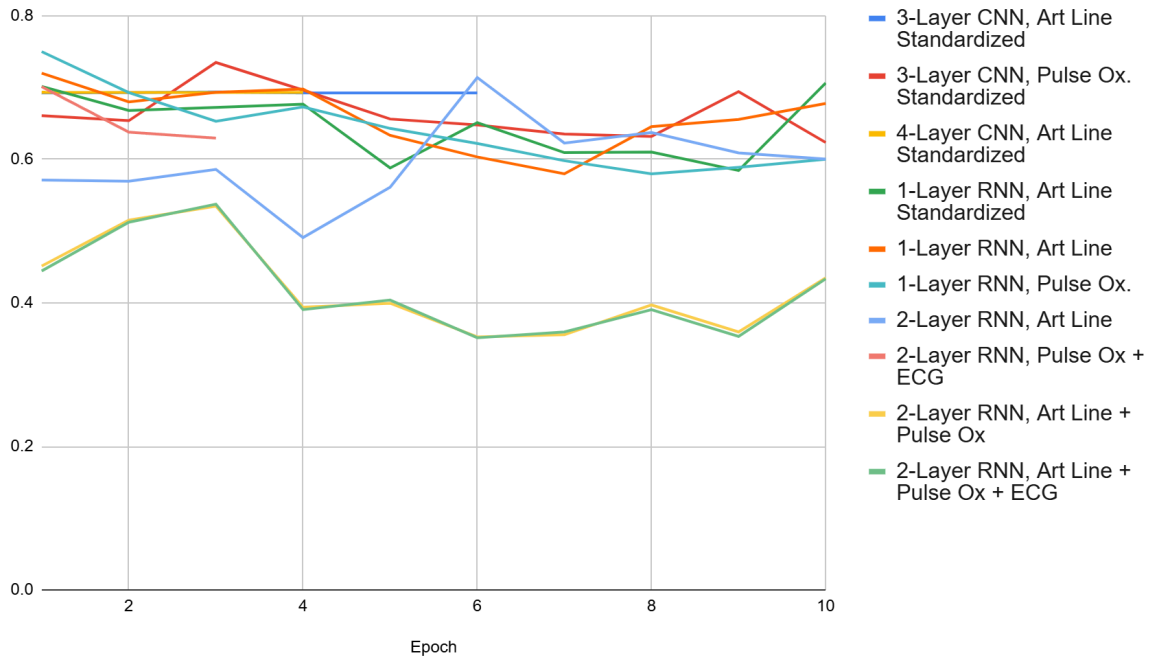


Figure C.4: Validation Accuracy vs. Epoch

